Lab 00 - Building the Home Lab & Portfolio Infrastructure - Read Me

> **Role:** Independent IT Professional **Difficulty:** ⭐☆☆☆☆ (long, not hard) **Time to Complete:** 5–7 hours (recommended: split across 2 sessions)

TL;DR

> Before solving any IT problems, I designed and built the entire infrastructure from scratch — a virtualized network with five machines running on VirtualBox, a functioning osTicket help desk system deployed on a LAMP stack, and a documentation pipeline across GitHub and Hashnode. Every lab going forward is logged as a real ticket with SLA tracking, troubleshooting documentation, and STAR method write-ups.

Documentation

Document	Description
📋 README.md	This file — overview, objectives, skills
📖 Full Lab	Complete walkthrough with screenshots and explanations
⚡ Condensed Lab	Quick reference runbook — steps only
🌐 Network Documentation	IP assignments and network diagram

Why This Lab Exists

The most common question I see in IT communities is: **"Where can I find labs for my portfolio?"**

I couldn't find what I was looking for — labs that simulate real-world incidents, document work through a ticketing system, and produce portfolio-ready output. So I built my own environment to make that possible.

Lab 00 is the foundation. Nothing else works without this.

What Was Built

```
YOUR ALIENWARE 16 AURORA (Host)
│
├── VirtualBox 7.x (Hypervisor)
│   ├── NAT Network: "LabNAT" (10.0.2.0/24 — internet access)
│   ├── Host-Only Network: "LabLAN" (192.168.56.0/24 — internal)
│   │
│   ├── [VM] osTicket-Server (Ubuntu Server 22.04)
│   │   └── Apache + MySQL + PHP + osTicket
│   │   └── IP: 192.168.56.10
│   │   └── Status: RUNNING
│   │
│   ├── [VM] Win10-PC01 (Shell — empty, configured, not installed)
│   │   └── IP: 192.168.56.100 (future)
│   │
│   ├── [VM] Win11-PC01 (Shell — empty, configured, not installed)
│   │   └── IP: 192.168.56.101 (future)
│   │
│   ├── [VM] Ubuntu-PC01 (Shell — empty, configured, not installed)
│   │   └── IP: 192.168.56.102 (future)
│   │
│   └── [VM] Mint-PC01 (Shell — empty, configured, not installed)
│       └── IP: 192.168.56.103 (future)
│
├── Documentation (your choice of platform)
│   └── STAR method writeups for each lab
│
└── ISOs Folder
    ├── Windows 10 Enterprise Evaluation
    ├── Windows 11 Enterprise Evaluation
    ├── Ubuntu Desktop 22.04.x LTS
    ├── Linux Mint 21.x
    └── Ubuntu Server 22.04.x LTS
```

Objectives

- [x] Install and configure **VirtualBox 7.x** with Extension Pack
- [x] Design and implement **dual-network architecture** (NAT + Host-Only)
- [x] Create **5 virtual machines** (1 server + 4 client shells)
- [x] Deploy **Ubuntu Server 22.04** with static IP configuration
- [x] Install and configure a full **LAMP stack** (Linux, Apache, MySQL, PHP)
- [x] Deploy and customize **osTicket** help desk ticketing system

- [x] Configure **SLA tiers, departments, help topics, and support teams**
- [x] Download and organize **5 operating system ISOs**
- [x] Create **VM snapshot** for clean-state recovery
- [x] Document **network architecture** and **credential management**
- [x] Establish a **documentation method** using the STAR framework

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Environment Specs

Component	Details
Host Machine	Alienware 16 Aurora · 32GB RAM · 1TB Storage
Hypervisor	Oracle VirtualBox 7.x with Extension Pack
Lab Network	192.168.56.0/24 (Host-Only)
Internet Access	10.0.2.0/24 (NAT per VM)
Server	Ubuntu Server 22.04 · osTicket · LAMP Stack
osTicket URL	[**http://192.168.56.10/osticket/scp/**](http://192.168.56.10/osticket/scp/)

Network Architecture

IP Address Assignments

Hostname	Role	IP Address	OS	Status
Host	Physical Machine	192.168.56.1	Windows	✅ Active
osTicket-Server	Help Desk Server	192.168.56.10	Ubuntu Server 22.04	✅ Active
Win10-PC01	Client Workstation	192.168.56.100	Windows 10 Enterprise	⬜ Shell
Win11-PC01	Client Workstation	192.168.56.101	Windows 11 Enterprise	⬜ Shell
Ubuntu-PC01	Client Workstation	192.168.56.102	Ubuntu Desktop 22.04	⬜ Shell

Mint-PC01	Client Workstation	192.168.56.103	Linux Mint 21	⬜ Shell

Network Diagram

text

```
          ┌──────────────────┐
          │   INTERNET   │
          └──────────────────┘
                  │
      ┌──────────────────────────┐
      │  NAT (10.0.2.0/24)   │  ← Each VM gets its own NAT
      │  Gateway: 10.0.2.2   │    for internet access
      └──────────────────────────┘
                  │
  ┌────┬────┬────┬
```

Adapter 2	Host-Only (192.168.56.10 static)
Services	Apache, MySQL, PHP, osTicket
Snapshot	"Clean-osTicket-Install" saved

Win10-PC01 (Shell)

Setting	Value
OS	Windows 10 Enterprise (64-bit)
RAM	4096 MB (4 GB)
CPU	2
Disk	50 GB (dynamic)
Adapter 1	NAT
Adapter 2	Host-Only
Status	Configured, OS not installed (Lab 02)

Win11-PC01 (Shell)

Setting	Value
OS	Windows 11 Enterprise (64-bit)
RAM	4096 MB (4 GB)
CPU	2
Disk	64 GB (dynamic)
Adapter 1	NAT
Adapter 2	Host-Only
EFI	✅ Enabled
TPM	✅ 2.0
Status	Configured, OS not installed (Lab 02)

Ubuntu-PC01 (Shell)

Setting	Value
OS	Ubuntu Desktop 22.04 (64-bit)
RAM	4096 MB (4 GB)
CPU	2
Disk	25 GB (dynamic)
Adapter 1	NAT
Adapter 2	Host-Only
Status	Configured, OS not installed (Lab 02)

Mint-PC01 (Shell)

Setting	Value
OS	Linux Mint 21 (Ubuntu 64-bit)
RAM	2048 MB (2 GB)
CPU	2
Disk	25 GB (dynamic)
Adapter 1	NAT
Adapter 2	Host-Only
Status	Configured, OS not installed (Lab 02)

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osTicket Configuration

SLA Plans

Name	Grace Period	Schedule
SEV-A (Critical)	1 hour	24/7
SEV-B (High)	4 hours	24/5
SEV-C (Normal)	8 hours	Monday–Friday 8am–5pm

SEV-D (Low)	24 hours	Monday–Friday 8am–5pm

Departments

Department	Type
IT Support	Public
System Administration	Public
Networking	Public

Help Topics

Help Topic	Department	SLA
Password Reset	IT Support	SEV-D
Hardware Issue	IT Support	SEV-C
Software Issue	IT Support	SEV-C
Network Connectivity	Networking	SEV-B
Access / Permissions	IT Support	SEV-C
Workstation Not Booting	IT Support	SEV-B
Malware / Security Incident	System Administration	SEV-A
New Employee Setup	IT Support	SEV-C
Printer Issue	IT Support	SEV-D
General Inquiry	IT Support	SEV-D

Teams

Team	Purpose
Tier I Support	First-line response and triage
Tier II Support	Escalation and advanced troubleshooting

What I Learned

Building this lab isn't complicated, but it is labor intensive. The VirtualBox setup and VM shells took about an hour. The Ubuntu Server, LAMP stack, and osTicket configuration took the rest — easily 4-5 hours.

Key takeaways:

- **YAML is unforgiving.** The Netplan configuration for static IP assignment requires exact spacing — one wrong indent and the entire network config fails. This taught me to respect configuration file syntax and always test after applying changes.
- **Snapshots save hours.** Taking a clean snapshot after configuring osTicket means I can break things in future labs and roll back in seconds instead of rebuilding from scratch.
- **Documentation is the real skill.** Setting up VirtualBox is straightforward. Writing about it clearly enough that someone else could follow? That's the hard part — and the part employers actually care about.
- **The ticketing system changes everything.** Having osTicket forces me to document every lab like a real technician — opening tickets, writing resolution notes, tracking SLAs. This habit is more valuable than any single technical skill.

Skills Demonstrated

Skill	What I Did
System Configuration	Configured VirtualBox preferences, VM hardware specs, EFI/TPM settings
Local Area Networks	Designed and implemented virtual network with NAT + Host-Only adapters, assigned static IPs via Netplan
Data Storage Technologies	Created dynamically allocated virtual hard disks, planned storage allocation across 5 VMs
Command-Line Interface	Configured Ubuntu Server entirely via CLI — installed LAMP stack, edited YAML configs, managed services with systemctl
Information Systems Security	Secured MySQL installation, set Apache file permissions, established credential management practices
System Support	Built Linux server from scratch, installed and configured web application (osTicket) on LAMP stack
Documentation	Created network diagrams, IP tables, credential reference sheets, established STAR method as standard lab documentation framework

Job Role Applicability

🟢 Help Desk Tier 1

Understanding how a ticketing system is structured — SLAs, departments, help topics, escalation tiers — gives you context for how your tickets are prioritized and routed. Most Tier 1 techs use ticketing systems daily but never see how they're configured. You now understand both sides.

🔵 IT Support Specialist

Building a virtual lab environment demonstrates that you can set up and manage infrastructure independently. Deploying a LAMP stack, configuring static IPs, and managing services via CLI are daily tasks for IT support roles beyond basic help desk.

🟡 Desktop Support Technician

Managing multiple operating systems (Windows 10, Windows 11, Ubuntu, Linux Mint) across a network is core desktop support work. Understanding VirtualBox networking modes prepares you for working with any virtualization platform in production.

🟠 System Support Specialist

Deploying a web application on a Linux server, configuring Apache, securing MySQL, managing firewall rules, and planning network architecture are system administration fundamentals. This lab demonstrates you can build infrastructure, not just use it.

Tools & Technologies Used

Tool	Purpose
Oracle VirtualBox 7.x	Type 2 hypervisor for VM management
VirtualBox Extension Pack	USB 2.0/3.0, disk encryption, PXE boot support
Ubuntu Server 22.04 LTS	Server OS for hosting osTicket
Apache 2	Web server
MySQL	Database server for osTicket
PHP 8.1	Server-side scripting for osTicket
osTicket	Open-source help desk ticketing system
Netplan	Ubuntu network configuration (YAML-based)

File Structure

text
Lab-00/

```
├── README.md
├── screenshots/
│   ├── virtualbox-manager.png
│   ├── extension-pack-installed.png
│   ├── host-only-network-config.png
│   ├── dhcp-server-settings.png
│   ├── ipconfig-host-only.png
│   ├── netplan-config.png
│   ├── static-ip-configured.png
│   ├── ping-host-to-server.png
│   ├── apache-default-page.png
│   ├── osticket-install-success.png
│   ├── osticket-staff-panel.png
│   ├── osticket-help-topics.png
│   ├── osticket-sla-plans.png
│   ├── vm-snapshot-created.png
│   ├── all-vms-listed.png
│   └── isos-folder.png
└── documentation/
    └── network-documentation.md
```

What You'd Tell an Employer

Before starting any labs, I designed and built the entire infrastructure from scratch — a virtualized network with five machines and a help desk ticketing system running on a LAMP stack. Every lab I complete is logged as a real ticket with SLA tracking, troubleshooting documentation, and measurable outcomes."

What's Next

Lab 00 is the foundation. The environment is built, the ticketing

system is configured, and the documentation pipeline is live.

In **Lab 01**, I step into the role of a technician at NovaBridge

Technologies, a managed service provider. The team needs bootable

USB toolkits for new client onboarding — and when one of them

won't boot at a client site, I diagnose why.

➡️ **Lab 01: Creating Bootable USB Media**